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MILITARY SPECIFICATION

# BOILERS, HEATING (10,000,000 BTU/H MAXIMUM)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers self-contained, low-pressure boiler units, complete with firing or burner equipment and suitable for steam or hot water heating systems.

* 1.2 Classification. Boilers shall be of the following types, classes, and styles, as specified (see 6.2).

Type I - Firetube (horizontal return tubular and horizontal fire box).
Type II - Watertube (steel-tube, copper-tube).
Type III - Cast iron, sectional.

Class 1 - Steam.
Class 2 - Hot water.

Style B - Coal, automatic, stoker fired.
Style C - Oil, automatic-fired.
Style D - Gas, automatic-fired.
Style G - Oil and gas combination, automatic-fired.
Style H - Oil, automatic-fired, convertible to solid fuel firing; gross output capacity 2,000,000 British thermal units per hour (Btu/h) or greater.
Style I - Gas, automatic-fired, convertible to solid fuel firing; gross output capacity 2,000,000 Btu/h or greater.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4410

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## 2. APPLICABLE DOCUMENTS

### 2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-P-28 - Paint; Aluminum, Heat Resisting (1,200°).
VV-F-815 - Fuel Oil, Burner.

MILITARY

MIL-V-173 - Varnish, Moisture and Fungus Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-B-3180 - Boiler and Related Equipment, Packaging of.
MIL-T-13867 - Treatment, Moisture and Fungus Resistant for Fire Control Electrical and Electronic Instruments and Equipment.
MIL-B-18796 - Burner, Single, Oil, Gas, and Gas Oil Combination (400,000 Btu's Per Hour and Over Input Capacity).
MIL-B-18797 - Burners, Oil, Mechanical Draft, Automatic (Under 400,000 Btu/hr Input Capacity).
MIL-I-24092 - Insulating Varnish, Electrical, Impregnating, Solvent Containing.

STANDARDS

FEDERAL

FED-STD-595 - Colors.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptability Requirements for the Control of Electromagnetic Interference.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)



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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN BOILER MANUFACTURERS ASSOCIATION (ABMA)

Packaged Firetube Boiler Engineering Manual.

(Application for copies should be addressed to the American Boiler Manufacturers Association, Suite 700, AM Building, 1500 Wilson Blvd., Arlington, VA 22209.)

AMERICAN GAS ASSOCIATION LABORATORIES (AGA)

Directory of Certified Appliances and Accessories.

(Application for copies should be addressed to the American Gas Association Laboratories, 1515 Wilson Blvd., Arlington, VA 22209.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

Z21.13 - Gas-Fired Steam and Hot Water Boilers.
Z83.3 - Gas Utilization Equipment in Large Boilers.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IV, Heating Boilers.
Performance Test Code.
PTC 4.1 - Steam Generating Units.

(Application for copies should be addressed to the American Society of Mechanical Engineers, P.O. Box 3199, Grand Central Station, New York, NY 10163.)

HYDRONICS INSTITUTE

(Institute of Boiler and Radiator Manufacturers Div.), (IBR)
(Steel Boiler Industry Div.), (SBI)

IBR - Testing and Rating Code for Low Pressure Cast Iron Heating Boilers.
SBI - Testing and Rating Code for Steel Boilers.

(Application for copies should be addressed to the Hydronics Institute, 35 Russo Place, Berkeley Heights, NJ 07922.)

NATIONAL BOARD OF BOILER AND PRESSURE VESSEL INSPECTORS (NBBPVI)

Relieving Capacities of Safety Valves and Relief Valves.

(Application for copies should be addressed to the National Board of Boiler and Pressure Vessel Inspectors, 1155 North High Street, Columbus, OH 43201.)



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NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ICS - 1 - General Standards for Industrial Controls and Systems.
ICS - 2 - Industrial Control Devices, Controllers and Assemblies.
ICS - 6 - Enclosures for Industrial Controls and Systems.
MG - 1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 70 - National Electrical Code.
No. 85 - Prevention of Furnace Explosions in Fuel Oil- and Natural Gas-fired Single Burner Boiler-Furnaces.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC., (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

UNDERWRITERS' LABORATORIES, INC. (UL)

UL296 - Oil Burners.
UL726 - Oil-fired Boiler Assemblies.
UL795 - Commercial-Industrial Gas-Heating Equipment.
Electrical Construction Materials List.
Gas and Oil Equipment List.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

## 3. REQUIREMENTS

3.1 Description. Boilers shall be type I, II, or III, as specified (see 6.2). All boilers except styles H and I shall be furnished complete with firing equipment, combustion chamber, insulation with steel jacket, safety and operating controls, integral electrical wiring, and other appurtenances, to make the boiler a complete, self-contained, fully automatic unit, ready for service upon completion of utility connections. Styles H and I boilers shall be furnished complete with firing equipment, combustion chamber, and safety and operating controls. Firing equipment and safety and operating controls may be packaged separately from the boiler. Insulation, if not furnished with the boiler, may be applied in the field. Nomenclature of ASME Boiler Pressure Vessel Code, Section IV, Heating Boilers shall apply.



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3.2 Design. The boilers shall be designed in accordance with the applicable sections of the ASME, Boiler and Pressure Vessel Code, Section IV, Heating Boilers and any additional requirements specified herein. The equipment design and accessory installations shall permit accessibility for maintenance and service in the field. Drainage and washout openings, when applicable, shall permit the boiler to be drained.

3.2.1 Design conditions. Class 1 boilers shall be designed for operation at pressures not exceeding 15 pounds per square inch gage (psig). Class 2 boilers shall be designed and constructed for operation at maximum pressures as specified (see 6.2 and 6.4) and stamped for the allowable working pressure in accordance with ASME Boiler and Pressure Vessel Code Section IV, Heating Boilers. For class 2 boilers, operating pressures shall be not less than 30 psig and shall not exceed 160 psig and temperatures shall not exceed 250° Fahrenheit (F) at or near the boiler outlet.

(a) Feedwater temperature.
(b) Boiler site elevation, if over 2,000 feet altitude.
(c) Maximum and minimum anticipated ambient air temperature.
(d) Maximum continuous gross output in British thermal units per hour (Btu/h) (see 3.6).

* 3.3 Standard commercial product. Each boiler of the same classification shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the boiler being furnished. A standard commercial product is a product which has been sold or is currently being offered for sale, on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

* 3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.5 Interchangeability. All boilers of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

* 3.6 Performance. When fired by the fuel specified in 3.9.1, 3.9.2, or 3.9.3, boiler capacity shall be as specified (see 6.2), expressed in gross output in Btu/h. When capacity ratings are given in pounds of steam per hour, one pound of steam per hour shall equal 970 Btu/h (the heat required to change water at 212°F to saturated steam at 212°F at standard conditions).



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* 3.6.1 Efficiency. Overall efficiency, based on boiler input and output capacities, determined as specified in 4.4.3, shall not be less than that indicated in table I for rated loads.

TABLE I. Overall boiler efficiencies.

| Fuel | | Gross output capacity up to 400,000 Btu/h | Gross output capacity over 400,000 Btu/h |
|---|---|---|---|
| Oil | Forced draft (power burner) | 77 percent | 82 percent |
| Gas | Forced draft (power burner) | 75 percent | 80 percent |
| | Natural Draft (atmospheric burner) | 75 percent | 80 percent |
| Coal | Mechanical Fired Forced and induced | 75 percent | 75 percent |

* 3.6.2 Heat input and heat transfer rates. For boilers having AGA or HI certification, the heat input and heat transfer rates, when the boiler operates at maximum gross output, shall be specified according to the testing and rating code of the AGA or HI. Unless otherwise specified (see 6.2), for other boilers the heat input and heat transfer rates, when the boiler operates at maximum gross output, shall not exceed the rates listed in table II.

TABLE II. Heat input and heat transfer rates.

| Type boiler | Heat input per unit volume (see 6.1.2.1) Btu/h/cu ft | Heat transfer per net of effective unit heating surface based on gross output (see 6.1.2.2) Btu/h/sq ft |
|---|---|---|
| Steel fire tube | 64,000 | 6,700 |
| Watertube (bent tube or serpentine) | 64,000 | 8,700 |
| Watertube (straight) | 64,000 | 6,700 |
| Cast iron | 64,000 | 6,700 |

3.6.2.1 Heat input. For purposes of this specification, the heat input shall be indicated in terms of Btu/h of the higher heating value of the fuel, and shall be calculated in accordance with the following formula:

$$\text{Heat input (Btu/h)} = \frac{\text{Gross output (Btu/h)}}{\text{Overall efficiency (as decimal)}}$$

* 3.6.2.2 Combustion exit temperature. The combustion exit temperature at the exit of the combustion chamber during maximum output shall not exceed 100°F below the initial deformation temperature for fusibility when coal-fired with a maximum grate temperature of 600°F. The minimum stack temperature shall be 100°F above the dew point of the combustion gas, but shall not exceed 600°F.

3.7 Construction. The boiler shall be constructed in accordance with all applicable requirements of the ASME Boiler and Pressure Vessel Code, Section IV, Heating Boilers. Steam boilers shall be so constructed as to insure that the moisture content of the steam leaving the boiler shall not exceed 2 percent with boiler water concentrations not in excess of those specified in the ABMA



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Packaged Firetube Boiler Engineering Manual. All necessary access doors and clean-out openings shall be furnished to provide full access to all fireside surfaces for inspection, cleaning, and repair. Observation openings shall be furnished to provide for visual inspection of combustion conditions. All doors, covers, and observation ports shall be reasonably gastight under all conditions of firing. The combustion setting, walls, and floor shall be constructed in accordance with commercial design utilized by the manufacturer and shall permit free expansion of the boiler without placing undue stress on any part of the boiler setting or tubes and tube sheets. All refractories shall be of the quality used in commercial practice for the required service and shall be capable of withstanding temperatures present under maximum load conditions. The refractory shall be readily replaceable on the job and shall be designed and secured in position in a manner to withstand jolts and shocks in commercial shipment. Component parts of the boiler assembly shall be arranged with respect to each other to permit easy access for servicing or repair and replacement of parts. Boilers shall have adequate stay bolts, tie rods, expansion provisions, and other design features required to prevent damage. The combustion chamber volume, combustion chamber height, and grate area for coal-fired boilers shall be not less than the minimum specified in the IBR or SBI Testing and Rating Code.

3.8 Boiler trim. Boiler trim and accessories shall comply with functional and installation requirements of ASME Boiler and Pressure Vessel Code, Section IV, Heating Boilers.

3.8.1 Class 1 boilers. All steam boilers shall be provided with the following boiler trim and accessories:

(a) One or more safety valves(s) (listed in NBBVI booklet, "Relieving Capacities as Safety Valves and Relief Valves").
(b) A steam pressure gage and a vacuum gage (or a compound steam pressure and vacuum gage).
(c) An adjustable pressure-combustion control to limit steam pressure up to 15 psig maximum.
(d) Bottom blowoff valve tapping and valve.
(e) One or more water gage set(s) with integral drain cocks.
(f) A water column assembly, except for boilers under 400,000 Btu/h which have gage glass connections provided on the boiler shell. The water column assembly shall be provided with a drain cock.
(g) Two automatic low-water level cutoff switches, one to be float actuated, the other to be float actuated or of the electrode type, installed as recommended in Section IV of the ASME, Boiler and Pressure Vessel Code. Low water shall activate a separate red indicator light.
(h) An automatic feedwater device to maintain the lowest safe water level.
(i) For steam boilers rated less than 400,000 Btu/h input, an automatic, combination low-water cutoff and feedwater device may be provided as an alternative to items (g) and (h).

3.8.2 Class 2 boilers. All hot water boilers shall be provided with the following trim and accessories:



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(a) One or more relief valves (listed in NBBPVI booklet, "Relieving Capacities of Safety Valves and Relief Valves").
(b) Pressure and altitude gages or combination pressure and altitude gage.
(c) A thermometer to indicate water temperature.
(d) A combination gage that comprises the thermometer, pressure, and altitude gages (may be provided as an alternative to items (b) and (c)).
(e) An adjustable temperature combustion control to limit water temperature to not more than 250°F.
(f) Bottom drain valve tapping.
(g) For hot water boilers rated 400,000 Btu/h or greater, an automatic low-water cutoff switch or water feeding device or both installed as recommended in Section IV of the ASME, Boiler and Pressure Vessel Code.
(h) For coil type or watertube hot water boilers rated 400,000 Btu/h or greater, a flow-sensing device in lieu of the low-water cutoff switch.

3.9.3 Hot water coils. When specified (see 6.2), boilers shall be equipped with internal hot water coils to serve as instantaneous heaters for service water. The coils shall be finned seamless copper not less than 3/4 inch in outside diameter. The coils shall terminate in threaded inlet and outlet connections on the exterior of the boiler, and shall be designed for the specific temperature rise and maximum draw rate, as specified (see 6.2). Unless otherwise specified (see 6.2), the pressure drop through the coil at the maximum temperature and draw rate shall not exceed 20 pounds per square inch (psi).

3.9 Firing equipment. Unless otherwise specified (see 6.2), firing and burner equipment shall be supplied with each boiler in accordance with the following applicable subparagraphs.

* 3.9.1 Coal-fired units. Units to be coal-fired shall be designed to operate with the type and grade of coal specified (see 6.2). The units shall be provided with automatic stokers of the covered hopper or bin feed type, as specified (see 6.2). Unless definite limits are specified (see 6.2), combustion chamber and grate heat releases shall be in accordance with the IBR or SBI Testing and Rating Code.

* 3.9.1.1 Coal feed systems. The coal feed system shall be of the screw feed or plunger feed type, as specified (see 6.2). The screw feed transmission shall be of the continuous feed mechanical type, totally enclosed in an oiltight casing and shall have either gear, worm and gear reduction, or double ratchet drive. The mechanical plunger feed shall consist of a connecting rod driven by a heavy-duty crank connecting plunger and pusher bar to the drive.

* 3.9.1.2 Retorts. Retorts shall be of the dead plate type, of cast iron sectional construction or enclosed windbox retorts with or without dead plates.

* 3.9.1.3 Tuyeres. Tuyeres shall be of heat-resisting cast iron with removable sections and shall have self-cleaning air openings. Tuyeres shall be accurately sized and proportioned for the specified coal and burning rates.



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* 3.9.1.4 Grates. Both dead and dump plates shall be of heavy cast iron, ribbed, finned or channeled on their undersurface contacting the cool air travel within the windbox with smooth surface to the fire side. Grates may be of the stationary, dump, or step type.

* 3.9.1.5 Draft fan. The draft fan shall be for a pressurized or balanced draft system, as specified (see 6.2); shall be integrally or separately mounted, as specified (see 6.2); and shall be electric-motor-driven for variable or fixed speed; direct or V-belt drive, as specified (see 6.2). Bearings for integral mounted fans shall be of the type normally furnished by the manufacturer. Bearings for separate mounted fans shall be self-lubricating, antifriction of the self-alining sheave type or roller bearings mounted in suitable pillow blocks, as specified (see 6.2).

* 3.9.1.5.1 Pressurized draft system. Adequate corrections shall be included in the fan selection or design for maximum burner pressure drop, combustion chamber pressure loss, combustion air temperature, and plant elevation to give a static pressure available at the flue stack outlet of the boiler or hot water generator in inches of water, as specified (see 6.2). Centrifugal fans shall have backward curved blades for the pressurized draft system.

* 3.9.1.5.2 Balanced draft system. The balanced draft system shall have a forced draft fan and an induced draft fan described herein. The static pressure required at the forced draft fan outlet shall be as specified (see 6.2), plus the maximum stoker pressure drop to give a positive or balanced (zero) pressure as specified (see 6.2), at the inlet of the stoker. Centrifugal fans shall have backward curved blades for the forced draft fan. The inlet and outlet static pressure required at the induced draft fan shall be as specified (see 6.2). The induced draft fan shall be designed for handling hot combustion flue gases at a maximum temperature, as specified (see 6.2), and shall have water cooled bearings when operating over 600°F. Centrifugal fans for induced draft service shall have radial-tip or forward-blade design.

* 3.9.1.6 Automatic ash removal. When specified (see 6.2), ash shall be removed from the boiler setting to ash collection point by automatic ash removal equipment of the feed screw scraper or vacuum type. Automatic ash removal equipment shall be complete with feedscrews, vacuum fans, driving motors, transmission equipment, controls, and necessary accessory equipment.

3.9.2 Style C and H units. The oil burners shall conform to the requirements of the specifications listed in table III. Burners and fuel systems shall be designed to fire fuel oil conforming to the requirements of VV-F-815, with grade and supply temperature as specified (see 6.2).

TABLE III. Specification requirements for style C and H burners.

| Input capacity Btu/h | Document number |
|---|---|
| Less than 400,000 | UL296 and MIL-B-18797 |
| 400,000 and over | UL296 and MIL-B-18796 |



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3.9.3 Style D and I units. Gas-fired units shall be designed for automatic operation and shall conform to the requirements of the specifications listed in table IV. Burners shall be designed for use with one or more of the following types of gases, as specified (see 6.2), having a specific gravity, heating value, and available pressure at the burner, as specified (see 6.2):

(a) Natural gas.
(b) Manufactured gas.
(c) Mixed gas.
(d) Liquified petroleum gas.
(e) Liquified petroleum gas-air mixture.

TABLE IV. Specification requirements for style D and I burners.

| Input capacity Btu/h | Document number |
|---|---|
| Less than 400,000 | ANSI Z21.13 |
| 400,000 to 6,000,000 | MIL-B-18796 and ANSI Z21.13 or UL795 |
| Over 6,000,000 | MIL-B-18796 |

3.9.4 Style G units. Oil and gas combination burners shall conform to the requirements of the specifications listed in table V. Burners and fuel systems shall be designed to fire fuel oil conforming to the requirements of VV-F-815, with grade and supply temperature as specified (see 6.2). Gas burners shall be designed for use with one or more of the following types of fuel, as specified (see 6.2). Fuel shall have a specific gravity, heating value, and available pressure at the burner, as specified (see 6.2):

(a) Natural gas.
(b) Manufactured gas.
(c) Mixed gas.
(d) Liquified petroleum gas.
(e) Liquified petroleum gas-air mixture.

TABLE V. Specification requirements for style G burners.

| Input capacity Btu/h | Document number |
|---|---|
| Less than 400,000 | Not permitted |
| 400,000 and over | MIL-B-18796 |

3.10 Controls. Pressure and temperature limit controls shall be adjustable to permit the selection of a predetermined operating steam pressure or water temperature limit setting, as specified (see 6.2), but in no case shall they be capable of being set in excess of the pressure or the temperature limits specified in 3.2.1. Pressure limit controls for steam boilers shall function as a pressure-actuated electrical switch. Temperature limit controls shall be of the thermostatic immersion type and shall be located in close proximity to the hottest portion of the boiler water space. Burners having input capacities less than 400,000 Btu/h shall have ON-OFF combustion control. Except for burners rated under 400,000 Btu/h, controls shall be designed for integrally mounting at one side of the boiler unless free standing control panels are required.



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3.10.1 Controls for style B units. Automatic stoker units shall be suitable for intermittent ON-OFF operation under thermostatic control. A stoker relay shall be provided to actuate the stoker to "full on" (maximum fuel rate) firing on thermostatic demand for heat, and also to provide minimum (pilot) fire when the thermostatic demand for heat is satisfied. Boilers with output capacities of 150 boiler horsepower or over shall be equipped with power-operated bypass type dampers for draft control. The thermostatic control shall include adjustable operating switches to permit selection of operating differential setting. Means shall be incorporated in the stoker control mechanism to stop the feeding of fuel if the fire goes out. When required by the particular stoker maintain proper draft conditions during intermittent operation of the stoker and when wind or weather would cause variable stack draft. A manual cutoff switch shall be provided to permit shutting off the stoker to effect repairs, without disconnecting the wiring.

3.10.2 Controls for style C and H units. All operating and safety controls shall be provided, as required, by the applicable specification covering the burner as provided by table III. Barometric-type dampers shall be provided for oil burning units with input capacities up to 400,000 Btu/h.

3.10.3 Controls for style D units. Controls for gas-fired units shall be as required by the burner specifications referenced in table IV, as applicable.

3.10.4 Controls for style G and I units. Controls for style G and I units shall meet the requirements of the burner documents referenced in table V.

3.11 Insulation and enclosure. Boilers shall be provided with an insulated sheet steel enclosure with a corrosion-resistant finish on both inside and outside surfaces. Insulation shall be noncombustible, nonhydroscopic, nonpowdering, and shall have the thickness and insulating qualities required so that the average surface temperature of the casing shall not exceed 150°F in an ambient temperature of 100°F with a surface wind velocity of 2 feet per second while the boiler is operated at full rated load. The insulation shall be mechanically supported other than by steel casing. Nominal thickness of the sheet steel enclosure shall be .0254 inch (U.S. revised standard gage No. 22). Enclosures shall be of sturdy construction, suitable for disassembly, and provide ready access to controls of firing accessories and equipment enclosed therein. When specified (see 6.2), firing equipment for style C and D units shall also be enclosed.

* 3.12 Electrical wiring and equipment.

* 3.12.1 Electric motors. Electric motors for the driven auxiliaries shall be constructed and rated in accordance with the applicable provisions of NEMA MG-1. Motors shall be general purpose, continuous duty, ball-bearing, induction motors. The motors shall be designed for operation on a power supply having characteristics as specified (see 6.2). Unless otherwise specified (see 6.2), motors shall be furnished with sealed dripproof or totally enclosed fan-cooled enclosures and windings.

* 3.12.2 Motor starters. Unless otherwise specified (see 6.2), motors shall be provided with magnetic across-the-line starters with overload and undervoltage protection and manual-off-automatic control stations. Magnetic starters shall conform to the applicable requirements of NEMA ICS-1, ICS-2, and ICS-6, and shall have NEMA type 12 enclosures.



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* 3.12.3 Wiring and circuit safety devices. All electrical wiring and circuit safety devices shall conform to NFPA No. 70. All control circuits shall be two pole, three wire with ground, not to exceed 120 volts nominal, and be protected with suitable fuses or circuit breakers.

* 3.12.4 Circuit breaker. A separate circuit breaker shall be provided for the steam generator.

* 3.13 Electromagnetic interference suppression. When specified (see 6.2), the boiler shall conform to the electromagnetic interference suppression requirements of MIL-STD-461, class C3, group I.

* 3.14 Cleaning, treatment, and painting. Unless otherwise specified (see 6.2), surfaces normally painted shall be cleaned, treated, and painted in accordance with the manufacturer's standard practice as long as the process meets the criteria specified herein. The color of the finish coat except for surfaces to be painted aluminum shall be in accordance with FED-STD-595 number as specified (see 6.2). Component parts such as motors, etc., procured factory finished need not be repainted. The paint shall be free from runs, sags, orange peel, or other defects.

* 3.14.1 Cleaning. Metal surfaces to be painted shall be cleaned to insure that they are free from all oil, grease, welding slag and spatter, mill scale, products of corrosion, dirt or other foreign substances. When aluminum heat-resisting paint is required (see 3.14.4), all exterior surfaces of the boiler shall be cleaned by commercial brush-off blast prior to assembly. Mechanical cleaning need not be performed on sheet steel jacket with straw colored mill scale, if the surfaces are free of all loose mill scale and products of combustion.

* 3.14.2 Treatment. All metal surfaces, except for units to be painted with aluminum heat-resisting paint, shall be treated to increase adherence of the coating system. A primer pretreatment coating shall be applied after cleaning and before any evidence of rust or other contamination can result.

* 3.14.3 Manufacturer's standard painting. The manufacturer's standard materials and standard processes shall be used to apply paint consisting of not less than one coat of rust-inhibiting primer and one coat of finish enamel with a total dry film thickness of not less than 2.5 mils. If the paint is electrostatically applied, it shall consist of not less than one coat of finish enamel with an average film thickness of not less than 2.5 mils.

* 3.14.4 Aluminum heat-resisting painting. When specified (see 6.2), before assembly, the boiler shell and the interior of the jacket shall be finished with one coat of aluminum heat-resisting paint conforming to TT-P-28. The paint shall be applied directly to clean bare metal surfaces and shall attain a minimum dry film thickness of 1 mil. After assembly, all exposed surfaces of the boiler and associated equipment normally painted in commercial practice shall be cleaned as specified herein and finished with two coats of the aluminum heat-resisting paint. Each coat shall be applied to a minimum dry thickness of 1 mil.



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* 3.15 Lubrication. Unless otherwise specified (see 6.2), means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. All parts requiring lubrication shall be properly lubricated before delivery and shall be tagged in a conspicuous place to indicate the temperature range and grade of lubricant used. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high pressure lubricating equipment, 1,000 psi or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location.

* 3.16 Fungus resistance. When specified (see 6.2), electric components and circuit elements including terminal and circuit connections shall be treated for moisture and fungus resistance in accordance with MIL-T-13867 as specified herein. Electrical components with a temperature rise of 87°F or less shall be coated in accordance with Type II of MIL-T-13867 with fungus resistance varnish conforming to MIL-V-173. Components with a temperature rise exceeding 87°F shall be coated in accordance with Type III or IV of MIL-T-13867 with varnish conforming to MIL-I-24092 with grade and class of varnish dependent on the component, class of insulation above 87°F and the recommendations of the manufacturer of the equipment.

3.17 Lifting attachments. Suitable lifting attachments and tiedown devices shall be furnished in accordance with the manufacturer's standard practice except as specified herein. The attachments and tiedown devices shall have eyes not less than 3 inches in inside diameter, and plates to withstand any handling conditions encountered. The tiedown devices shall be located at each corner of the boiler. Adequate tiedown devices may also be used for lifting purposes. Each attachment or tiedown device shall be of sufficient capacity to carry the total weight.

3.18 Identification plate. The contracting officer will furnish to the Government inspector, the required identification plates. The contractor will be required to stamp the necessary data in the blank spaces thereon and securely affix said plates in a conspicuous place on each unit, assembly and/or subassembly, and parts as directed by the Government inspector. Nonferrous screws, rivets, or bolts of not less than 1/8 inch in diameter shall be used to affix the plates.

3.19 Identification marking. Equipment, assemblies, and parts shall be marked for identification in accordance with the manufacturer's standard practice.

3.20 Technical publications. Unless otherwise specified (see 6.2 and 6.3), one commercial manual shall be furnished with each boiler. The manual shall include the operation, inspection, and maintenance instructions recommended in UL296, 726, 795, NFPA No. 85, ANSI Z21.13, and Z83.3, as applicable.

3.21 Maintenance tools. A suitable flue-cleaning brush or scraper shall be furnished with each coal-fired boiler.

3.22 Workmanship and fabrication. Workmanship and fabrication shall be in accordance with the ASME Boiler and Pressure Vessel Code, Section IV, and shall be of such quality as to produce boilers that meet the standards prevalent among manufacturers who normally produce equipment of the type specified herein.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Quality conformance inspection (see 4.2.1).
(b) On-site inspection (at site of installation) (see 4.2.2).

4.2.1 Quality conformance inspection. Quality conformance inspection shall be performed on all boilers offered to the Government. Inspection of the first unit shall include the examination of 4.3 and all the tests of 4.4. For subsequent units, the inspection shall consist of the examination of 4.3 and the tests of 4.4.1, 4.4.2.1, 4.4.2.2, and 4.4.4.

4.2.3 On-site inspection. The Government reserves the right to conduct, at the site of installation, field tests that may be deemed necessary by the procuring activity to verify compliance with the requirements of this specification. The contractor shall have the privilege of representation when such field tests are conducted. Field tests shall be conducted at Government expense. Deficiencies noted by such tests shall be corrected at the contractor's expense and any required retesting shall be at the contractor's expense. Field tests will be performed prior to putting the boiler in service, and within one year from date of delivery to the Government.

4.3 Examination. Each boiler shall be examined to verify compliance with the requirements specified in Section 3 of this specification. The examination shall be as specified in table IV. Any boiler containing one or more defects shall be rejected.

TABLE IV. Classification of defects.

| Categories | Defects |
|---|---|
| Critical: | |
| 01 | Programming combustion and safety control systems including flame safeguard control, boiler limit controls, and fuel safety interlocks not as specified (see 3.1 and 3.10). |
| 02 | Number of safety valves and aggregate relieving capacity not in accordance with ASME Boiler and Pressure Vessel Code (see 3.2, 3.7, 3.8.1, and 3.8.2). |



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TABLE IV. Classification of defects. (Cont'd)

| Categories | Defects |
|---|---|
| Major: | |
| 101 | Boilers not of the type, class or style specified (see 1.2). |
| 102 | Boiler design, construction, and materials not in accordance with the applicable provisions of this specification and the ASME Boiler and Pressure Vessel Code, Section IV (see 3.2, 3.3, 3.4, and 3.7). |
| 103 | Insufficient: (1) combustion chamber volume, (2) effective radiant heating surface, (3) or total effective heating surface for burner input, as applicable (see 3.6.2 and 4.3.1). |
| 104 | Boiler not complete. Component parts including boiler trim, fittings, or accessories missing (see 3.7, 3.8, 3.17, and 3.18). |
| 105 | Automatic low-water cutoff and feed water regulator not furnished or as specified (see 3.8.1 and 3.8.2). |
| 106 | Wiring not in accordance with NFPA No. 70 (see 3.12). |
| 107 | Treatment and painting not as specified (see 3.14). |
| 108 | Workmanship and fabrication not as specified (see 3.23). |
| Minor: | |
| 201 | Markings not as specified (see 3.17, 3.18, and 3.20). |
| 202 | Specified technical publications missing or incomplete (see 3.21). |
| 203 | Specified tools missing or incomplete (see 3.22). |

4.3.1 Furnace heat rates. When specified (see 6.2 and 6.3), as part of the first unit production, the manufacturer shall furnish data sheets including required calculations and sketches to indicate conformance with 3.6.2. The data sheets should show the heating surface as indicated in 6.1.2.1. Combustion chamber dimensions should also be given to indicate conformance with 6.1.2.2.

4.4 Tests. Each boiler shall be tested in accordance with the test requirements of 4.2. Any unit failing to pass the following tests, as applicable, shall be cause for rejection. Certified test reports (see 6.3) from an independent laboratory approved by the contracting officer will be accepted as evidence that the unit has passed the required tests. Tests shall be conducted as outlined in the referenced documents as specified herein.

4.4.1 Fire and explosion safety tests. Tests shall be conducted as necessary to determine compliance with the applicable UL Safety Standards or ANSI Standards. Acceptable evidence of compliance with requirements of these standards shall be the applicable UL listing mark, inclusion in the UL Gas and Oil Equipment List, or certification symbol of the AGA Laboratories or a listing in the AGA Directory of Certified Appliances and Accessories.

4.4.1.1 Oil-fired boilers. Oil-fired boilers shall meet the test requirements of UL726. Burners on boiler-burner units listed as approved under UL726 will be considered as burners that meet the requirements of UL296.

4.4.1.2 Oil burners. Oil burners shall meet the test requirements of UL296.



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4.4.1.3 Gas-fired boiler and gas burners. Gas-fired boilers and gas burners shall meet the test requirements of UL795, ANSI Z21.13, or ANSI Z83.3, as applicable.

4.4.2 Strength and tightness tests. The boiler(s) shall be subjected to the following strength and tightness tests.

4.4.2.1 Boiler shell including fittings and accessories. The boiler shall be hydrostatically tested in accordance with the requirements of the ASME Boiler and Pressure Vessel Code, Section IV. The ASME stamp symbol will be accepted as evidence of this test.

4.4.2.2 Fuel system. The part of the preassembled fuel oil system that is furnished integrally with the boiler shall be hydrostatically tested at 1-1/2 times the maximum operating pressure. The part of the preassembled gas system that is furnished integrally with the boiler shall be pneumatically tested at operating pressure. The soap bubble or the halogen-sniff test method shall be used to verify tightness of the gas system. Acceptable evidence of compliance with the test requirements shall be the UL listing mark or the AGA Laboratories Certification.

4.4.3 Capacity and efficiency tests. The specified capacity, efficiency, and steam quality of the boiler shall be determined in accordance with one of the following test procedures, as applicable:

(a) ASME PTC 4.1 short form.
(b) The IBR Testing and Rating Code for Low Pressure Cast Iron Heating Boilers.
(c) The SBI Testing and Rating Code for Steel Boilers.
(d) ANSI Z21.13.
(e) ANSI Z83.3.

Efficiency shall be determined and performance shall be as specified in 3.6. The IBR emblem, the SBI symbol, the HI certification symbol, or the AGA Laboratories Certification Symbol, as applicable, will be acceptable in lieu of capacity and efficiency tests when accompanied by a certificate of compliance as part of first unit of production (see 6.4), which states that the specified boiler capacity and efficiency were determined by the physical test procedures outlined in the applicable rating code or requirements of ANSI Standards.

4.4.4 Operational tests. Unless otherwise specified (see 6.2), the boiler shall be tested to demonstrate control and operational conformance with the requirements of this specification and the supplementary burner specifications. Oil and gas burners shall be tested for ability to start a cold boiler in accordance with the manufacturer's instructions. Unless otherwise specified (see 6.2), oil burners shall be tested for ability to make a cold start from the specified oil supply temperature of 3.9.2. Oil and gas burners shall also be tested to determine whether they provide steam or hot water in accordance with steam or hot water demand for ON-OFF cycling. Boilers that exhibit: (1) excessive or unexplained loss of ignition, (2) nuisance shutdowns due to faulty burner or control operation, (3) improper flame, (4) excessive carbon deposits, or (5) necessity for frequent or difficult adjustments shall be rejected.



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4.4.5 Electromagnetic interference suppression tests. When conformance with electromagnetic interference limits is required, the equipment shall be tested in accordance with MIL-STD-462 to determine conformance with electromagnetic interference suppression requirements of 3.13. When suppressed to conform to the electromagnetic interference requirements of 3.13, the manufacturer may upon approval of the contracting officer furnish a certification in lieu of the test (see 6.3), that the boilers meet the requirements, together with a list of the suppression devices installed. The list shall be sufficiently detailed to allow visual determination that the devices are installed.

4.5 Packaging inspection. The preservation-packaging, packing, and marking of the boiler shall be inspected to determine conformance to the applicable requirements of Section 5 of this specification.

5. PACKAGING

5.1 Preservation-packaging, packing, and marking. Unless otherwise specified (see 6.2), the boilers shall be preserved, packaged, packed, and marked in accordance with MIL-B-3180. The level of preservation and packaging and level of packing shall be as specified (see 6.2).

6. NOTES

6.1 Intended use. The equipment specified herein is intended for hot water heating or low-pressure steam heating systems.

6.1.1 Copper-tube boilers. Watertube boilers provided with copper tubes are intended for use with class 2, hot water boilers.

6.1.2 Definitions.

6.1.2.1 Combustion chamber volume. As used in table II, combustion chamber volume shall be defined as the cubic content of the space provided for combustion of the fuel. Combustion chamber limits shall be defined as not extending beyond the first plane of entry into, or between, the tubes. Screen tubes, when used, shall be considered as the first plane of entry.

6.1.2.2 Heating surface. As used in table II, the heating surface of boiler shall be computed in accordance with the ASME Boiler Pressure Vessel Code, Section IV, Heating Boilers.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, class, and style, and gross output capacity required (see 1.2, 3.1, and 3.6).
(c) When specifications and standards shall be other than as specified (see 2.1.1).
(d) Required allowable working pressure of class 2 boilers (hot water) (see 3.2.1).
(e) Design conditions, as applicable (see 3.2.1 and 3.6).



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(1) Feedwater temperature.
(2) Site elevation.
(3) Anticipated maximum and minimum ambient air temperatures.
(4) Maximum continuous gross output in Btu/h.

(f) Heat transfer ratio if different (see 3.6.2).
(g) When hot water coils are required; temperature rise and draw rate; pressure drop, if different (see 3.8.3).
(h) When firing and burner equipment is not to be provided (see 3.9).
(i) Type and grade of coal required; type of stoker required; if combustion chamber and grate heat releases are to be different than IBR and SBI Codes (see 3.9.1).
(j) Type of coal feed system required (see 3.9.1.1).
(k) Type of draft system required; required type of mounting for fan; fixed or variable speed; direct or V-belt electric-motor-driven; and type of bearings required for separate mounted fans (see 3.9.1.5).
(l) For pressurized draft system, static pressure required at the flue stack outlet of the boiler (see 3.9.1.5.1).
(m) For balanced draft system, static pressure required at the forced draft fan outlet; the positive or balanced (zero) static pressure at the inlet of the combustion chamber; the inlet and outlet static pressure required of the induced draft fan; and the maximum temperature of the flue gases (see 3.9.1.5.2).
(n) When automatic ash removal equipment is required (see 3.9.1.6).
(o) Fuel oil grade and supply temperature (see 3.9.2 and 3.9.4).
(p) Fuel gas characteristics (see 3.9.3 and 3.9.4).

(1) Type.
(2) Specific gravity.
(3) Heating value.
(4) Available pressure at the burner.

(q) Operating steam pressure or water temperature as applicable (see 3.10).
(r) When firing equipment for style C and D units are to be enclosed (see 3.11).
(s) Electrical power supply characteristics; when motors shall be different (see 3.12.1).
(t) When motor starters shall be different (see 3.12.2).
(u) When electromagnetic interference suppression is required (see 3.13).
(v) When painting is to be other than manufacturer's standard; color of finish coat required (see 3.14).
(w) When surfaces are to be painted with aluminum heat-resisting paint conforming to TT-P-28 (see 3.14.4).
(x) When lubrication shall not be in accordance with manufacturer's standard practice (see 3.15).
(y) When fungus resistance is required (see 3.16).
(z) Requirements of technical publications, if different (see 3.20 and 6.3).

(aa) When data sheets are required (see 4.3.1).



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(bb) When testing to demonstrate control and operational conformance shall be different (see 4.4.4).
(cc) When oil burners shall not be tested for their cold start ability at oil supply temperature (see 4.4.4).
(dd) If preservation-packaging, packing, and marking is other than specified (see 5.1).
(ee) Level of preservation-packaging and level of packing required (see 5.1).

6.3 Contract data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 Pressure limitation. The ASME Boiler and Pressure Vessel Code, Section IV, limits the operating pressure for class 1 boilers (steam) to 15 psig. Class 2 boilers (hot water) are usually designed for operating pressures of 30, 60, 100, 120, and 160 psig. The ASME Code, Section IV, limits hot water heating boilers to a maximum operating pressure of 160 psig and a boiler water temperature of 250°F.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Navy - YD
Air Force - 99

Review activities:

Army - CE, ME

User activity:

Navy - MC

Preparing activity:

Navy - YD

Project No. 4410-0048



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| | |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |
| | |

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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